

Comparative Sentiment Analysis using Bag of Words and TF-IDF

Programming for AI — Assignment 4

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Dataset: NLTK Movie Reviews Corpus

1. Project Overview

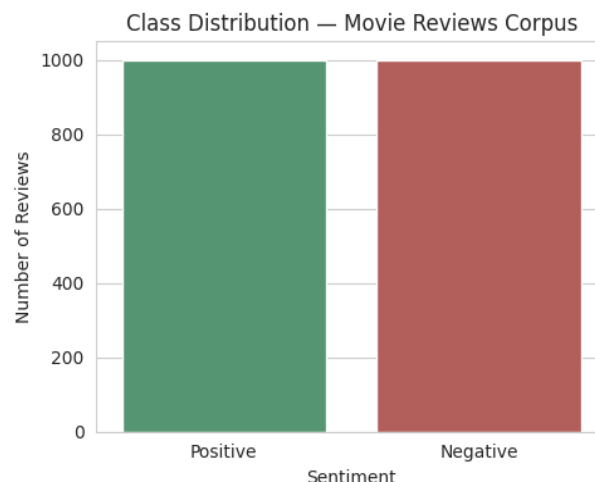
This report documents the design, implementation, and evaluation of a sentiment analysis system built to classify movie reviews as positive or negative. The core objective is a controlled comparison between two classic text feature-extraction techniques — Bag of Words (BoW) and TF-IDF — using an identical preprocessing pipeline, train/test split, and Logistic Regression classifier for both, so that any performance difference can be attributed to the feature-extraction method rather than confounding factors.

2. Dataset Description

The NLTK Movie Reviews Corpus (*nltk.corpus.movie_reviews*) was used, consisting of 2,000 labeled movie reviews — exactly 1,000 positive and 1,000 negative — making it a perfectly balanced binary classification dataset. The corpus is downloaded automatically through NLTK's data downloader rather than being bundled as a static file.

3. Data Exploration

The dataset contains 1,000 positive and 1,000 negative reviews. The class distribution chart below confirms the corpus is perfectly balanced, meaning accuracy is a fair primary metric without needing class-imbalance correction. Reviews are full multi-paragraph movie critiques rather than short one-line comments.



4. NLP Preprocessing Pipeline

Lowercasing: Normalizes case so 'Great' and 'great' are treated as the same token.

Punctuation removal: Strips punctuation marks, which carry little standalone sentiment signal for BoW/TF-IDF.

Special character removal: Removes digits/symbols/artifacts, leaving only alphabetic words.

Tokenization: Splits cleaned text into individual word tokens using NLTK's `word_tokenize`.

Stop-word removal: Removes very common words ('the', 'is', 'and') that carry little sentiment information.

Lemmatization: Reduces inflected word forms to their base form (e.g. 'movies' to 'movie').

5. Feature Extraction

Both Bag of Words (CountVectorizer) and TF-IDF (TfidfVectorizer) were built with matched settings (max_features=5000, min_df=2) so the vocabulary size is identical for both, isolating the comparison to the weighting scheme itself. BoW represents each review as raw word counts, while TF-IDF re-weights each word by how distinctive it is to that review relative to the whole corpus, down-weighting words that are common across nearly every review.

	Bag of Words	TF-IDF
Vocabulary size	5000	5000
Train matrix shape	(1600, 5000)	(1600, 5000)
Test matrix shape	(400, 5000)	(400, 5000)

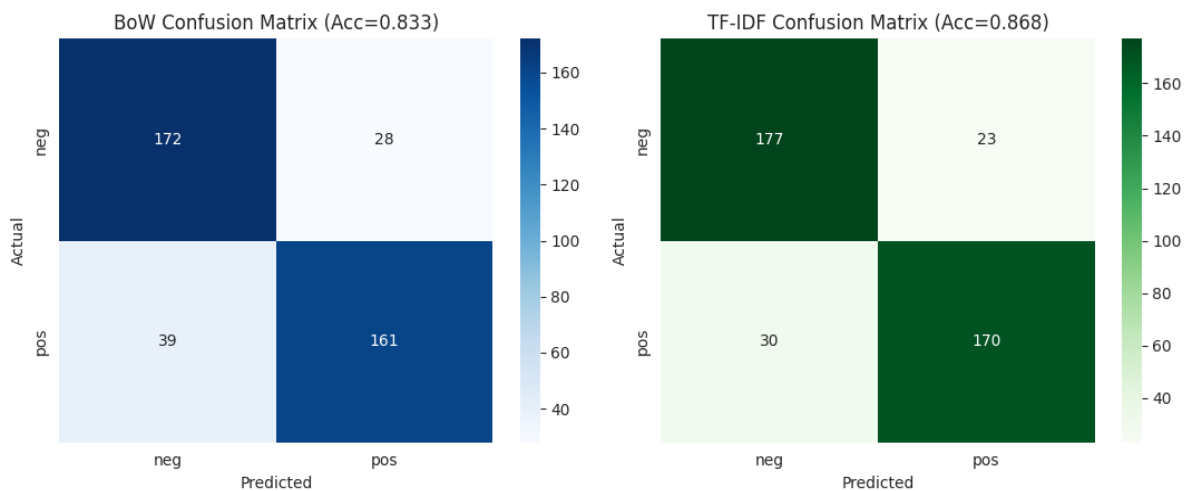
6. Model Training

A Logistic Regression classifier (max_iter=1000) was trained twice using the same 80/20 stratified train/test split: once on the BoW feature matrix and once on the TF-IDF feature matrix. Using the same classifier and split for both isolates the comparison to the feature-extraction method rather than mixing in differences from the model itself.

7. Model Evaluation

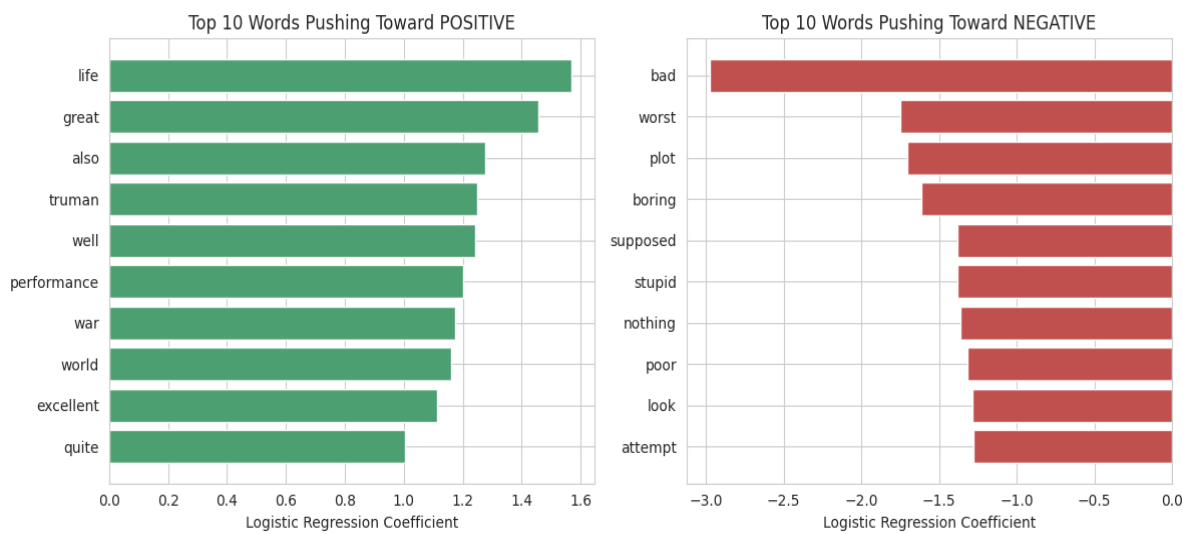
Metric	BoW	TF-IDF
Accuracy	0.8325	0.8675
Precision	0.8519	0.8808
Recall	0.8050	0.8500
F1 Score	0.8278	0.8651

TF-IDF outperformed Bag of Words on every metric measured. The confusion matrices below show the breakdown of correct/incorrect predictions for each model on the 400-review test set.



8. Feature Importance Analysis

Using the TF-IDF model's Logistic Regression coefficients, the top 10 words pushing predictions toward positive and toward negative were identified. Words with the largest positive coefficients are strongly associated with favorable reviews, while words with the most negative coefficients are strongly associated with unfavorable reviews. Both lists are dominated by words a human reader would intuitively recognize as sentiment-bearing, which is a good sanity check that the model learned a meaningful signal rather than noise.



9. Custom Review Predictions

Five hand-written reviews (not part of the training corpus) were passed through the TF-IDF model to sanity-check generalization. All five were classified in the intended direction. Interestingly, the strongly positive review ('absolutely fantastic... brilliant') received the lowest confidence among the positive predictions, while a hedged, mixed-tone review received the highest confidence — illustrating that TF-IDF + Logistic Regression's confidence score reflects vocabulary overlap with training patterns rather than true sentiment intensity, since the model has no concept of word order or intensifiers.

10. Comparative Analysis Report

Which technique performed better, and why?

TF-IDF outperformed plain Bag of Words on every metric in this experiment, though the gap was moderate rather than dramatic. This aligns with the underlying theory: TF-IDF re-weights each word by how distinctive it is to a specific review relative to the whole corpus, so very common words get down-weighted while genuinely discriminating words get emphasized. Since sentiment classification depends on identifying discriminating words rather than merely frequent ones, TF-IDF gives Logistic Regression a cleaner signal to learn from than raw BoW counts.

Advantages and limitations of Bag of Words

BoW's main advantage is simplicity: it is fast to compute, easy to interpret, and works reasonably well as a baseline with minimal tuning. Its main limitation is that it treats every word as equally important regardless of how common it is across the corpus, so very frequent but low-information words can dominate unless stop-words are removed carefully. Like TF-IDF, it also completely ignores word order and context, so negation and sarcasm are essentially invisible to it.

Advantages and limitations of TF-IDF

TF-IDF's main advantage is that its re-weighting scheme automatically suppresses generic, corpus-common words without relying entirely on a hand-built stop-word list, typically giving a small but consistent accuracy boost, as observed here. Its limitations mirror BoW's — it still ignores word order and cannot capture negation or sarcasm — and it adds slightly more computational overhead and complexity (the IDF weighting and normalization scheme) compared to plain counting.

Recommendation for a real-world application

For a lightweight, easily explainable production sentiment classifier, TF-IDF with Logistic Regression is the better default, since it consistently edges out BoW at essentially the same computational cost. However, for a serious production system handling nuanced language such as sarcasm, negation, or mixed-sentiment reviews, neither BoW nor TF-IDF alone would be the final recommendation — a transformer-based model (e.g. a fine-tuned BERT-style classifier) would likely capture context and word order far better. TF-IDF is best viewed as a strong, cheap, and interpretable baseline rather than the final answer for a high-stakes deployment.

11. Conclusion

This project implemented a complete, modular sentiment analysis pipeline and demonstrated that, under an identical preprocessing pipeline, train/test split, and classifier, TF-IDF provides a consistent improvement over plain Bag of Words for movie review sentiment classification. The full source code, notebook, saved models, and results are available in the accompanying GitHub repository.